

Date: 30/07/2025

Lab Practical #09:

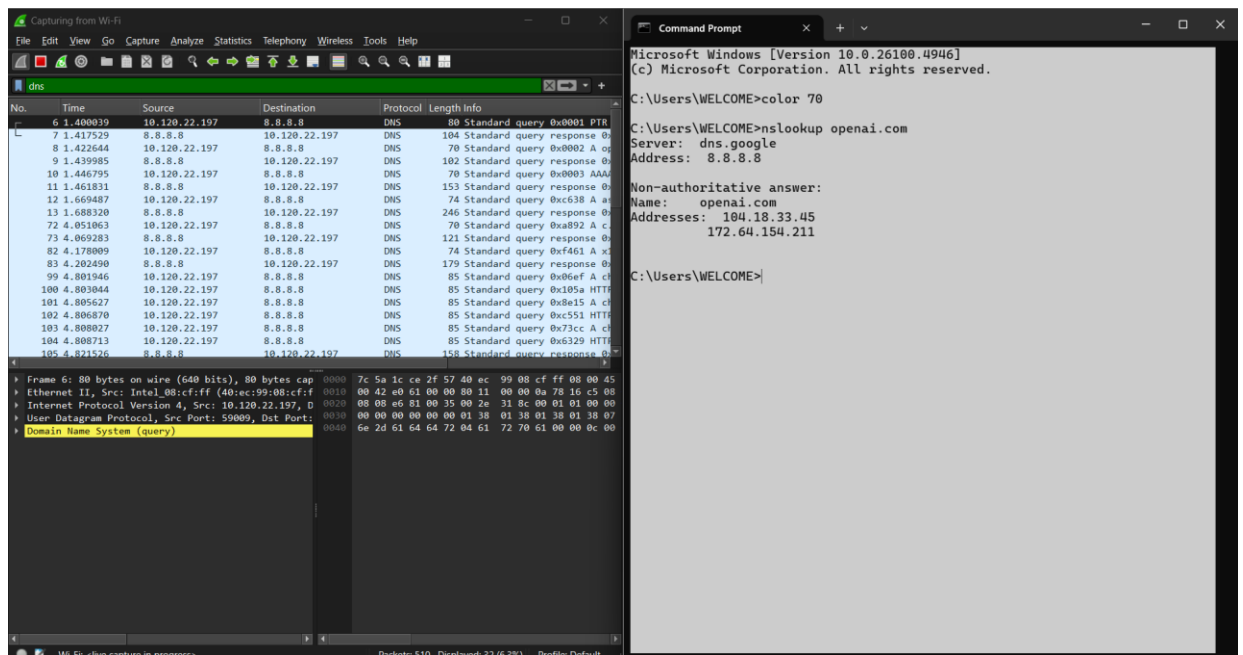
Study Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

Practical Assignment #09:

1. Explain usage of Wireshark tool.

Wireshark is a network protocol analyzer used to capture and analyze network traffic in real time. It helps in troubleshooting network issues, monitoring performance, detecting security threats, and learning protocol behavior.

2. Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)



The screenshot displays two windows side-by-side. The left window is Wireshark, showing a packet capture from a Wi-Fi interface. The packet list on the left shows several DNS queries and responses. The selected packet (No. 6) is a DNS query from 10.120.22.197 to 8.8.8.8. The packet details pane shows the 'Domain Name System (query)' section. The packet bytes pane shows the raw data in hexadecimal and ASCII. The right window is a Command Prompt, showing the execution of the 'nslookup openai.com' command, which returns the IP address 104.18.33.45.

No.	Time	Source	Destination	Protocol	Length	Info
6	1.400039	10.120.22.197	8.8.8.8	DNS	80	Standard query 0x0001 PTR
7	1.417529	8.8.8.8	10.120.22.197	DNS	104	Standard query response 0x0001 PTR
8	1.422644	10.120.22.197	8.8.8.8	DNS	70	Standard query 0x0002 A of
9	1.439985	8.8.8.8	10.120.22.197	DNS	102	Standard query response 0x0002 A of
10	1.446795	10.120.22.197	8.8.8.8	DNS	70	Standard query 0x0003 AAAA
11	1.461831	8.8.8.8	10.120.22.197	DNS	153	Standard query response 0x0003 AAAA
12	1.669487	10.120.22.197	8.8.8.8	DNS	74	Standard query 0xc638 A a
13	1.688320	8.8.8.8	10.120.22.197	DNS	246	Standard query response 0xc638 A a
72	4.051063	10.120.22.197	8.8.8.8	DNS	70	Standard query 0xa892 A c
73	4.069283	8.8.8.8	10.120.22.197	DNS	121	Standard query response 0xa892 A c
82	4.178009	10.120.22.197	8.8.8.8	DNS	74	Standard query 0xf461 A x
83	4.202490	8.8.8.8	10.120.22.197	DNS	179	Standard query response 0xf461 A x
99	4.801946	10.120.22.197	8.8.8.8	DNS	85	Standard query 0x06ef A c
100	4.803044	10.120.22.197	8.8.8.8	DNS	85	Standard query 0x105a HTTP
101	4.805627	10.120.22.197	8.8.8.8	DNS	85	Standard query 0x8e15 A c
102	4.806870	10.120.22.197	8.8.8.8	DNS	85	Standard query 0xc551 HTTP
103	4.808027	10.120.22.197	8.8.8.8	DNS	85	Standard query 0x73cc A c
104	4.808713	10.120.22.197	8.8.8.8	DNS	85	Standard query 0x6329 HTTP
105	4.821526	8.8.8.8	10.120.22.197	DNS	158	Standard query response 0x6329 HTTP

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Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

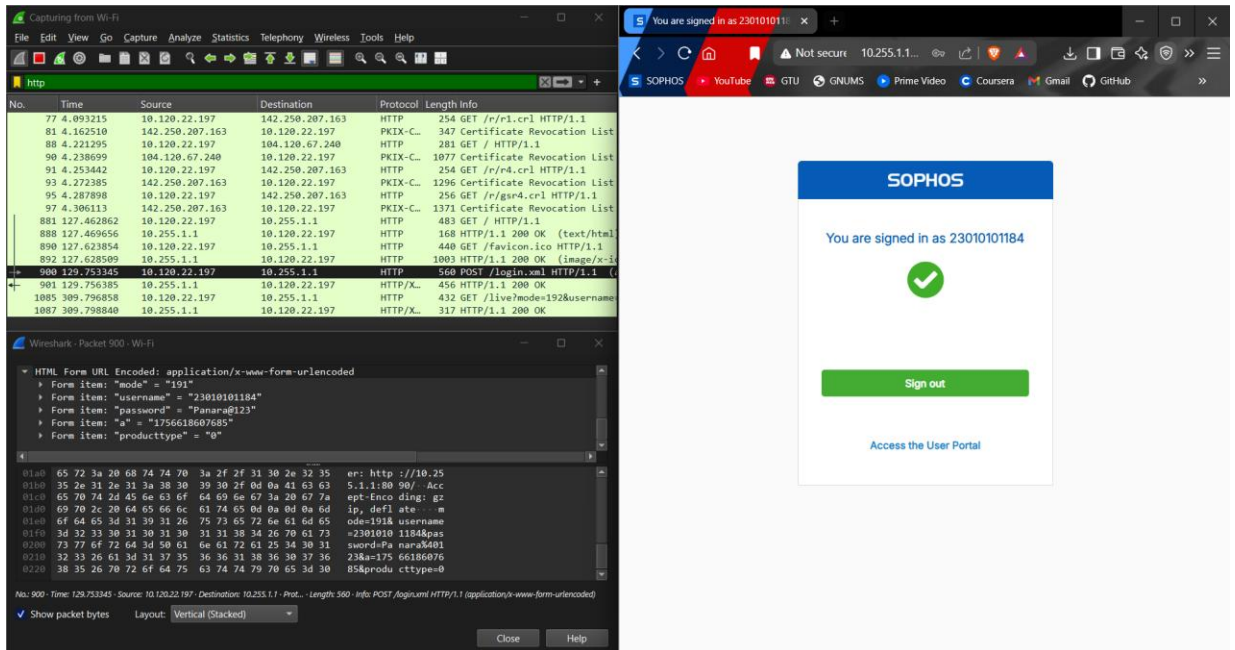
C:\Users\WELCOME>color 70

C:\Users\WELCOME>nslookup openai.com
Server:      dns.google
Address:     8.8.8.8

Non-authoritative answer:
Name:   openai.com
Address: 104.18.33.45
        172.64.154.211

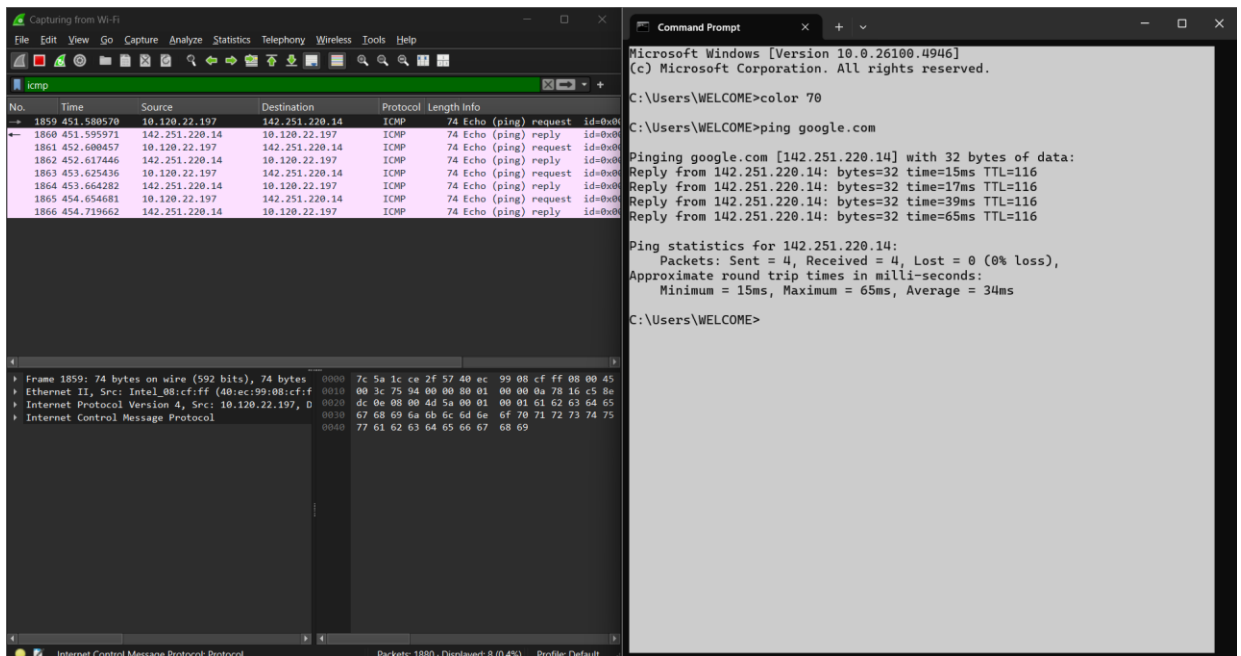
C:\Users\WELCOME>
```

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The top screenshot shows a Wireshark packet capture of an HTTP login attempt. The packet list shows a POST request to /login.xml. The packet details pane shows the HTML form data: mode=191, username=23010101184, password=Panara@123, and producttype=0. The packet bytes pane shows the raw data in hexadecimal and ASCII.

The bottom screenshot shows a web browser window displaying the Sophos login page. The page shows a success message: "You are signed in as 23010101184" with a green checkmark and a "Sign out" button. Below the message is a link to "Access the User Portal".



The top screenshot shows a Wireshark packet capture of an ICMP ping test. The packet list shows four echo requests and four echo replies. The packet details pane shows the ICMP Echo (ping) request and reply. The packet bytes pane shows the raw data in hexadecimal and ASCII.

The bottom screenshot shows a Windows Command Prompt window. The user has entered the command "color 70" and "ping google.com". The output shows the ping statistics for 142.251.220.14: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 15ms, Maximum = 65ms, Average = 34ms.



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 30/07/2025

The image shows a Wireshark packet capture window on the left and a web browser window on the right displaying the Darshan University Learning Management System (LMS) interface.

Wireshark Packet Capture:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.120.22.197	140.82.113.25	TCP	55	51287 → 443 [ACK] Seq=1 A
2	0.352348	140.82.113.25	10.120.22.197	TCP	66	443 → 51287 [ACK] Seq=1 A
3	1.243342	10.120.22.197	4.213.25.242	TLSv1.2	154	Application Data
4	1.267932	4.213.25.242	10.120.22.197	TLSv1.2	225	Application Data
5	1.321689	10.120.22.197	4.213.25.242	TCP	54	49758 → 443 [ACK] Seq=101
14	1.606954	10.120.22.197	23.58.95.136	TCP	66	51350 → 443 [SYN] Seq=0 W
15	1.792135	23.58.95.136	10.120.22.197	TCP	66	443 → 51350 [SYN, ACK] Seq=
16	1.792515	10.120.22.197	23.58.95.136	TCP	54	51350 → 443 [ACK] Seq=1 A
17	1.808442	10.120.22.197	23.58.95.136	TLSv1.3	837	Client Hello (SHA-assets.v
18	1.989025	23.58.95.136	10.120.22.197	TLSv1.3	318	Server Hello, Change Ciph
19	1.989217	10.120.22.197	23.58.95.136	TCP	54	51350 → 443 [ACK] Seq=784
20	2.020918	10.120.22.197	23.58.95.136	TLSv1.3	134	Change Cipher Spec, Appli
21	2.021767	10.120.22.197	23.58.95.136	TLSv1.3	134	Application Data
22	2.035216	10.120.22.197	23.58.95.136	TLSv1.3	2629	Application Data
23	2.117918	23.58.95.136	10.120.22.197	TCP	60	443 → 51350 [ACK] Seq=265
24	2.117918	23.58.95.136	10.120.22.197	TLSv1.3	357	Application Data
25	2.118126	10.120.22.197	23.58.95.136	TCP	54	51350 → 443 [ACK] Seq=351
26	2.118739	23.58.95.136	10.120.22.197	TCP	60	443 → 51350 [ACK] Seq=568
27	2.118739	23.58.95.136	10.120.22.197	TLSv1.3	115	Application Data

Web Browser (Darshan University LMS):

Semester 5

2301CS402 - Design and Analysis of Algorithm

2301CS501 - Computer Network

The image shows a Wireshark packet capture window displaying a detailed view of a TCP connection and TLS handshake.

Wireshark Packet Capture:

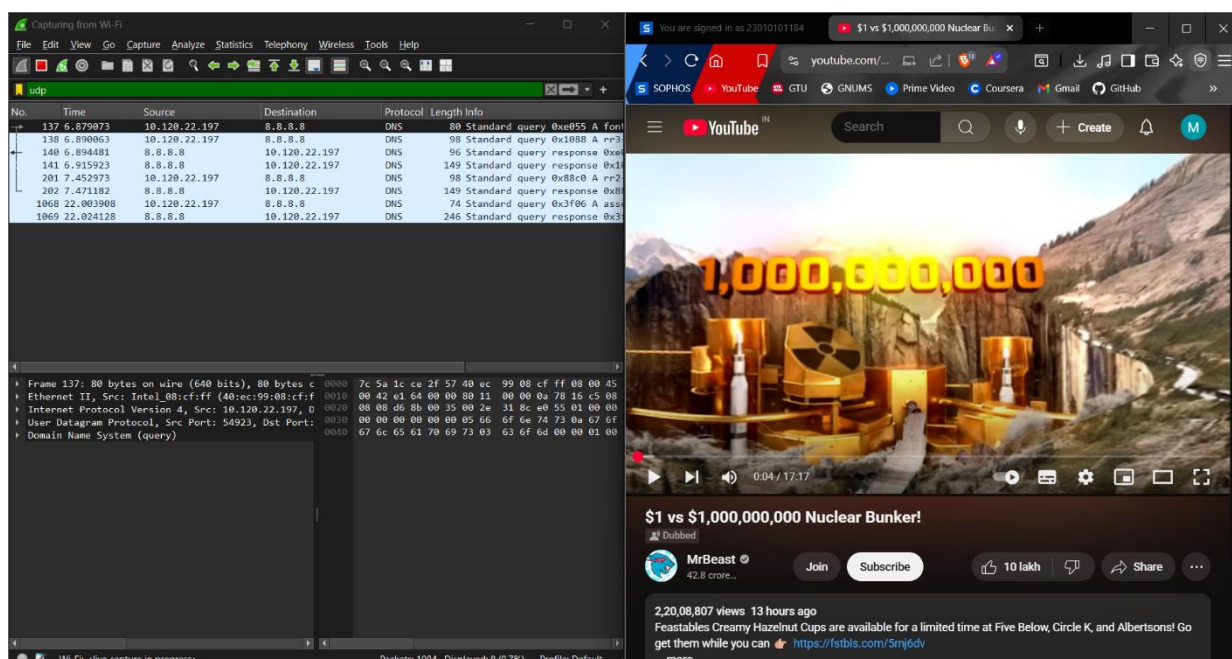
No.	Time	Source	Destination	Protocol	Length	Info
5378	112.488064	10.120.22.197	142.251.43.14	TLSv1.2	3383	Application Data
5379	112.494075	27.109.0.13	10.120.22.197	TLSv1.2	65006	Application Data, Application Data, Application Data, Application Data
5380	112.494075	27.109.0.13	10.120.22.197	TLSv1.2	29706	Application Data, Application Data
5381	112.494075	27.109.0.13	10.120.22.197	TLSv1.2	436	Application Data, Application Data, Application Data, Application Data, Application Data
5382	112.494064	10.120.22.197	27.109.0.13	TCP	54	51417 → 443 [ACK] Seq=400402 Ack=26490810 Win=2096096 Len=0
5383	112.508755	142.251.43.14	10.120.22.197	TCP	60	443 → 51405 [ACK] Seq=354595 Ack=174466 Win=1194 Len=0
5384	112.511932	142.251.43.14	10.120.22.197	TCP	60	443 → 51405 [ACK] Seq=354595 Ack=1747971 Win=1205 Len=0
5385	112.611909	142.251.43.14	10.120.22.197	TLSv1.2	124	Application Data
5386	112.614573	142.251.43.14	10.120.22.197	TLSv1.2	85	Application Data
5387	112.614573	142.251.43.14	10.120.22.197	TLSv1.2	93	Application Data
5388	112.614650	10.120.22.197	142.251.43.14	TCP	54	51405 → 443 [ACK] Seq=174971 Ack=354735 Win=508 Len=0
5389	112.615750	10.120.22.197	142.251.43.14	TLSv1.2	93	Application Data
5390	112.613864	142.251.43.14	10.120.22.197	TCP	60	443 → 51405 [ACK] Seq=354735 Ack=175010 Win=1205 Len=0
5391	121.145270	10.120.22.197	142.250.192.35	TCP	55	[TCP Keep-Alive] 51407 → 443 [ACK] Seq=215 Ack=146 Win=255 Len=1
5392	121.161335	142.250.192.35	10.120.22.197	TCP	66	[TCP Keep-Alive] 443 → 51407 [ACK] Seq=146 Ack=216 Win=1043 Len=0 SLE=215 SRE=216
5393	121.921446	10.120.22.197	142.251.43.1	TCP	55	[TCP Keep-Alive] 51410 → 443 [ACK] Seq=390 Ack=1946 Win=254 Len=1
5394	121.938880	142.251.43.1	10.120.22.197	TCP	66	[TCP Keep-Alive] 443 → 51410 [ACK] Seq=1946 Ack=391 Win=1043 Len=0 SLE=390 SRE=391
5395	122.310991	10.120.22.197	8.8.8.8	TCP	55	[TCP Keep-Alive] 51399 → 443 [ACK] Seq=498 Ack=1337 Win=250 Len=1
5396	122.329391	8.8.8.8	10.120.22.197	TCP	66	[TCP Keep-Alive] 443 → 51399 [ACK] Seq=1337 Ack=499 Win=1040 Len=0 SLE=408 SRE=499

Web Browser (Darshan University LMS):

Semester 5

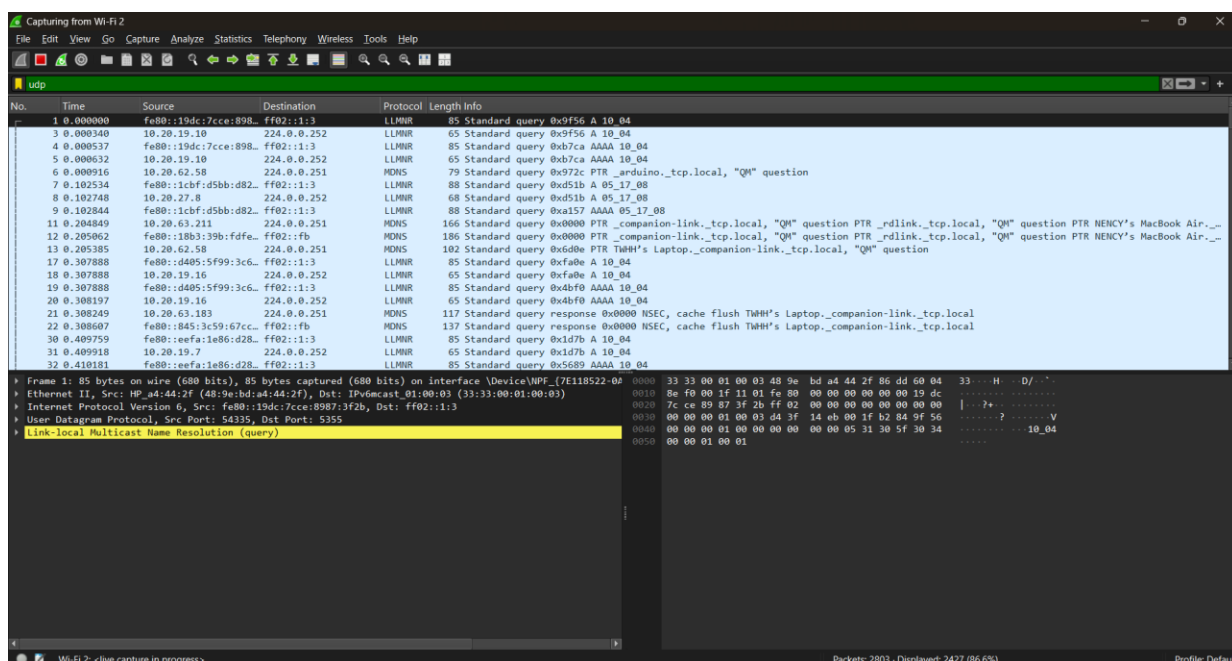
2301CS501 - Computer Network

Date: 30/07/2025



The image shows a Wireshark packet capture of a DNS query. The packet list shows a query for '0x855 A 10_04' from source 10.120.22.197 to destination 8.8.8.8. The packet details show the query structure. The packet bytes show the raw data. The packet list shows a response for '0x855 A 10_04' from source 8.8.8.8 to destination 10.120.22.197. The packet details show the response structure. The packet bytes show the raw data.

The image also shows a YouTube video player with the title '\$1 vs \$1,000,000,000 Nuclear Bunker!'. The video has 2,20,08,807 views and was uploaded 13 hours ago. The video content shows a nuclear bunker with a large '1,000,000,000' overlay.



The image shows a Wireshark packet capture of a DNS query. The packet list shows a query for '0x9F56 A 10_04' from source 10.20.19.10 to destination 224.0.0.252. The packet details show the query structure. The packet bytes show the raw data. The packet list shows a response for '0x9F56 A 10_04' from source 224.0.0.252 to destination 10.20.19.10. The packet details show the response structure. The packet bytes show the raw data.

The image also shows a YouTube video player with the title '\$1 vs \$1,000,000,000 Nuclear Bunker!'. The video has 2,20,08,807 views and was uploaded 13 hours ago. The video content shows a nuclear bunker with a large '1,000,000,000' overlay.